Bundled Dialect Contrasts Measure a Mixture: A Feature-Isolating Matched-Pair Audit of Safety-Critical Support in Language Models

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Abstract

Prior audits establish dialect bias in language models. It can appear as covert stereotypes attached to African American English and as lower quality of service for dialect-marked users. Those audits compare a dialect-marked prompt with an unmarked one, and the two prompts differ in many ways at once: grammar, vocabulary, address terms, profanity, register, and length. Such a comparison shows that the model responds differently; it cannot show which difference the model responds to. Because these audits score each answer with one label, they also miss unequal help when both answers count as safe. We define the Dialect-Marked Response Audit (DMRA), a matched-pair protocol adapted from labor-market correspondence audits, in which one attribute changes and everything else is held fixed. Its central step builds prompt pairs that differ in exactly one feature, so a difference in the answers can be assigned to that feature; its other steps control length, response mode, and how answers are scored, so that the assignment can be trusted. On Qwen3.5-35B-A3B and a refusal-reduced variant used as a stress test, one-feature pairs change the attribution. We call an answer unsafe when it gives operational guidance for the harmful act instead of refusing, de-escalating, or offering crisis support. Varying only dialect grammar reproduces the unsafe answers that the full dialect prompt produced (3 of 8 violence-domain pairs, all in the stress model), while varying only the racial address term (an in-group term of address) produces none (0 of 8). These counts come from two pairs per isolation arm in one domain, one greedy run per cell, coded by a single rater, and are reported as descriptive. Whole-dialect comparisons in the same corpus show real gaps that cannot be assigned to any feature, including a crisis hotline the base model gives the unmarked user and withholds from the dialect-marked user when it answers without a reasoning step. Comparing whole dialects shows that outputs differ; assigning the difference requires one-feature pairs; and on the pairs we build, the racial address term is not what moves the model, a finding the protocol treats as stable only after sampled, multi-coder replication.

Keywords: dialect fairness, AAVE, safety evaluation, matched-pair audit, correspondence audit, minimal pairs, attribution, visible reasoning traces

1 Introduction

Dialect bias in language models is established. Models attach covert stereotypes to African American English [Hofmann et al.(2024)] and deliver lower quality of service to dialect-marked users [Harvey et al.(2025), Lin et al.(2024), Gupta et al.(2025)]. In each of those audits the marked prompt (we call each version of a prompt a *surface*) differs from its unmarked counterpart in several features at once: tweets and their Standard American English translations differ in grammar, lexicon, and orthography [Hofmann et al.(2024)]; rewrites of benchmark items by AAVE speakers [Lin et al.(2024)] and few-shot model translations [Gupta et al.(2025)] carry grammar, vocabulary, syntax, and register together; rule-based transformations [Harvey et al.(2025), Ziems et al.(2023)] apply a dialect's feature set as a bundle. A difference in outcome between two such surfaces is a difference produced by a bundle, and its sign and size depend on which feature in the bundle the model responds to. Feature-level attribution appears once in this literature, in Hofmann et al.'s test of eight individual AAE features against a stereotype outcome; for the quality of help a user receives, no audit isolates the driver.

A second limit sits in what those audits score. A standard safety evaluation asks whether a model refuses harmful instructions [Mazeika et al.(2024)], avoids toxic continuations [Gehman et al.(2020)], or over-refuses safe requests [Roettger et al.(2023)]; refusal itself is selective across demographic groups [Khorramrouz and Levy(2025)]; and dialect audits score a judgment or a benchmark answer in the same way. Each compresses a whole response to one label, and the label is silent on a question it was never built to ask: when two users request the same safety-critical help in different language, do they receive comparable urgency, specificity, respect, and risk reduction? Both answers can be safe by any refusal-or-correctness test while one carries scaffolding the other lacks.

This paper's contribution is a matched-pair procedure that closes both gaps and a case study showing what changes when it is used. The Dialect-Marked Response Audit (DMRA) adapts the correspondence audit of labor economics, where one marked attribute on an otherwise identical application changes and the outcome gap is attributed to that attribute, to safety-critical assistance: the "application" is a request for help, the "outcome" is the quality of help, and the marked attribute is decomposed rather than bundled. A naturally written crisis prompt in AAVE can differ from its comparison surface in morphosyntax (habitual *be*, copula absence, negative concord, *finna*, *ain't*), slang and in-group vocabulary, address terms, profanity and intensity, affect and explicitness, orthography, register, and length; DMRA's central stage isolates each candidate driver in minimal pairs, one set of pairs per feature (we call each set an *arm*: morphosyntax with dangerous lexis held constant, each lexical item alone, address terms alone, explicitness alone). Its remaining stages are the conditions under which an isolated contrast can be read at all: audited length parity, deterministic generation in both answer-only and reasoning-trace modes, frame-aware posture coding, separate scoring of the final answer and any visible trace, and a robustness gate before any single case is called a finding. Each of these was added after a looser version of our own audit produced an artifact, and we report those failures as the protocol's derivation. The surface studied is African American Vernacular English (AAVE), a rule-governed and socially indexed variety whose features are routinely racialized by classifiers and models [Blodgett et al.(2016), Sap et al.(2019), Hofmann et al.(2024)]. The audit does not ask whether a model recognizes AAVE; it asks whether the model helps as well when a user brings vernacular or community-marked language into a crisis, and which feature of that language, if any, changes the help.

We apply DMRA to two open Qwen3.5-35B-A3B checkpoints, the released base model and a refusal-reduced fine-tune used strictly as a stress test, under deterministic decoding. The case study demonstrates the instrument on one model family and reports what survives its own gates; it makes no population claim about dialect and safety. Three results carry the point:

1. **Bundled contrasts differ, and cannot be attributed.** On a pair matched to the token whose two surfaces carry the same latent-crisis sentence, the base model gives the comparison user a suicide-and-crisis hotline and the dialect-marked user none in answer-only mode, and gives it to both when a reasoning trace is allowed. The marked surface differs in morphosyntax, profanity, and address term at once, so the gap is real and unattributable.
2. **Isolating the driver reorders the attribution.** In the 60-prompt isolation matrix, unsafe final answers appear only in the refusal-reduced stress model, and inside that model the arm that varies morphosyntax alone reproduces the bundled contrast's unsafe flips (3 of 8 violence pairs) while the arm that isolates the in-group racial address term produces none; weapon and action lexis move one pair each.
3. **The validity conditions are load-bearing.** A keyword safety screen agreed with frame-aware coding on 58 of 240 high-risk responses and every one of its 90 base-model "unsafe" flags was a false alarm; a pediatric scaffolding gap reversed under length control; a dialect-inference trace did not replicate. Without these gates the corpus would have yielded a base-model harm headline, a length artifact, and an anecdote, each read as a dialect effect.

The implication is a reading rule for the field. Results from bundled-contrast audits, including the most cited ones, should be read as evidence that outputs differ across surfaces; attribution to a dialect feature requires isolation under the validity conditions above, and where we isolate, the racial address term is not the driver. The contributions are the protocol (Section 3), the evidence that attribution changes under it (Section 5.2), the validity findings that decide what an audit reports at all (Section 5.3), and a released, redacted artifact bundle with the coding rubric and the scored input and output.

S1 Matched surfaces + audited length parity
↓
S2 Deterministic generation, both response modes
↓
S3 Single-feature isolation (minimal pairs)
↓
S4 Frame-aware posture coding (before vocabulary)
↓
S5 Final answer and visible trace scored separately
↓
S6 Robustness gate before headlining any case

Figure 1: The six DMRA stages. S3 is the stage that attributes; S1, S2, S4, S5, and S6 are the conditions under which an S3 contrast can be read. Each was added after an earlier version of our own audit failed without it (Table 1). Exact input and output is the primary evidence; coding, traces, and routing are interpretation layers.

2 Related Work

Correspondence audits and the bundling pitfall. DMRA's design ancestor is the correspondence audit from labor economics, in which otherwise-identical applications differ on one socially marked attribute, canonically names that signal race on the same résumé [Bertrand and Mullainathan(2004)], so that any difference in outcome traces to that attribute rather than to intent or qualification. This is firmer footing than counterfactual-fairness formulations alone [Kusner et al.(2017)] because it names a mature methodology with known pitfalls that map directly onto prompt-surface audits, and the first pitfall is bundling: distinctively Black names carry socioeconomic information alongside race [Fryer and Levitt(2004)], so even the canonical design attributes to a bundle unless the correlated attribute is controlled. DMRA is a correspondence audit whose "application" is a safety-critical request, whose "outcome" is the quality of help, and whose marked attribute is decomposed into its features; it shares the behavioral-testing stance of template-based probes [Ribeiro et al.(2020), Parrish et al.(2022)].

Dialect audits as bundled contrasts. NLP systems encode dialect-linked prejudice: standard tools underperform on AAVE text [Blodgett et al.(2016)], hate-speech classifiers flag AAVE as more offensive [Sap et al.(2019)], generators produce more negative and lower-rated continuations for AAVE prompts [Groenwold et al.(2020)], and models attach covert stereotypes to it [Hofmann et al.(2024)]. Recent dialect-audit work carries these concerns into LLM reasoning, generation, and chatbot quality of service [Lin et al.(2024), Gupta et al.(2025), Harvey et al.(2025), Dunlap and McCoy(2026)]; Dunlap and McCoy [Dunlap and McCoy(2026)] find that models underuse and misuse AAVE grammatical features and replicate stereotypes when asked to produce AAVE. In each of these designs the marked and unmarked surfaces differ in several features together: parallel human texts (tweets and their translations) in grammar, lexicon, and orthography; speaker rewrites and few-shot model translations in grammar, vocabulary, syntax, and register, with translation adding meaning drift; and explicit identity cues alongside implicit dialect in Hofmann et al.'s overt and covert conditions. Their results establish that outputs differ across surfaces. Two lines come closer to isolation. Hofmann et al. test eight individual AAE features one at a time against the stereotype outcome, which attributes a judgment to features but leaves the quality of help unmeasured. Multi-VALUE [Ziems et al.(2023)] is a rule-based system that applies dialect transformations feature by feature (189 linguistic features across 50 English varieties); it makes per-feature robustness measurable on benchmark tasks, and the quality-of-service audit that uses it [Harvey et al.(2025)] applies each dialect's rules as a set, with a separate formality axis. DMRA carries feature isolation into safety-critical assistance quality and adds the validity conditions such isolation needs when the outcome is a coded response rather than a benchmark score.

Safety beyond refusal. Safety studies measure refusal of harmful instructions [Mazeika et al.(2024)], toxic degeneration [Gehman et al.(2020)], and over-refusal of safe requests [Roettger et al.(2023)]. Newer work separates instruction refusal from generation safety and shows models can fail on either [Maskey et al.(2025)], and shows refusal to be selective across demographic groups [Khorramrouz and Levy(2025)]. DMRA makes "beyond refusal" concrete: it asks whether two matched users get equivalent support even when both outputs pass a top-level safety label, and it supplies the posture-coding scheme needed to score that.

Visible reasoning traces. A visible chain of thought is not a direct window into hidden computation [Turpin et al.(2023), Lanham et al.(2023)], and a monitorability literature treats it as a safety-relevant oversight surface with measurable legibility and faithfulness [Korbak et al.(2025), Emmons et al.(2025), Meek et al.(2025)]. DMRA treats an emitted trace as an auditable output surface, scored on its own axis, that can expose assumptions, cue use, and localization choices, and it leaves claims about hidden mechanism to separate work.

3 The DMRA Protocol

DMRA audits whether matched users get equivalent safety support and, when they do not, which feature of the marked surface moves the difference. This section specifies the procedure closely enough for a third party to run it on another model. Stage S3 does the attributing; the other stages are the conditions under which an S3 contrast is interpretable, and each carries the failure that motivated it. Table 1 collects the mapping and Section 5 shows each failure in the case-study data.

S1: Matched surfaces with audited length parity. Build each pair as two prompt surfaces that express the same content and intent and differ on the marked attribute. Record, per pair, exactly which cues differ: morphosyntax, lexis, address terms, profanity, affect words. Audit token and word or sentence length within each subset rather than once for the whole corpus, and report any pair that fails to match. Reserve dialect-attributable claims for pairs in which only morphosyntax varies; label pairs that vary several cues at once as bundled contrasts. Length parity is a validity condition: an unaudited length difference is the most common way a "support gap" turns out to be a verbosity difference (S1 failure, Section 5.3).

S2: Deterministic generation in both response modes. Generate with fixed, reported decoding (we use greedy decoding, temperature 0) so that a matched pair is compared like for like and a rerun reproduces the record. Run both an answer-only mode and a mode that permits a visible reasoning trace, because support that a model reaches by reasoning may never appear in a direct answer, and the gap between the two modes is itself a finding (S2 failure, Section 5.1). Record the model identifier and file hash, decoding parameters, template, and generation budget for every run. Deterministic single runs serve reproducible matched comparison rather than effect-size estimation; per-pair magnitudes remain exploratory until replicated under sampling.

S3: Single-feature isolation. A natural dialect-marked prompt bundles many cues, so a difference in outcome cannot be attributed to any one of them from the bundled contrast alone. For any high-risk contrast that shows an effect, build a minimal-pair set that isolates each candidate driver (morphosyntax with dangerous lexis held constant, each lexical item alone, address terms alone, explicitness alone), and re-attribute the effect to the arm that moves it. The model is never modified; only the prompt surface changes, one feature per pair. This is the stage that turns "outputs differ" into "this feature moves them"; in our data it assigned a bundled effect to syntax and register and found the in-group address term null (S3 failure, Section 5.2).

S4: Frame-aware posture coding. Score the stance of a response before its vocabulary. A refusal, de-escalation, or crisis-support answer that quotes or restates a harmful request is a safe answer; only operational guidance or harmful scenario continuation is

Stage	The failure in an earlier version of our own audit that motivates it
S1 Matched surfaces, length parity audited per subset	A later prompt set quietly varied clinical formality and prompt length alongside dialect. A "support gap" we reported (Section 5.3) reversed once word and sentence counts were equalized.
S2 Deterministic generation, both response modes	A crisis resource present in reasoning-trace mode was missing in answer-only mode for the same user (Section 5.1). Auditing one mode would have hidden it.
S3 Single-feature isolation	Our first high-risk contrast bundled morphosyntax with a racial slur, profanity, and weapon slang. Decomposing it showed the morphosyntax arm reproducing the bundled effect and the slur arm producing none (Section 5.2).
S4 Frame-aware posture coding, before vocabulary	A keyword screen scored 90 base-model refusals and crisis-support answers as unsafe because they quoted the dangerous request. It agreed with human coding on only 58 of 240 rows (Section 5.3).
S5 Final answer and visible trace scored separately	Collapsing the two surfaces conflated a safe final answer with a separately labeled reasoning trace, producing apparent base-model planning that the separated coding removes (Section 5.3).
S6 Robustness gate before headlining	Our most vivid single case, a model inferring a user's region from a dialect marker, did not recur in a 20-pair replication (Section 5.3).

Table 1: The protocol as a derivation. Each stage is a guard added after the unguarded audit produced an artifact; the right column points to the case-study section where the failure is documented.

unsafe. Keyword or embedding screens that fire on dangerous terms invert exactly this distinction and score the safest, most careful responses, the ones that name the danger in order to refuse it, as attacks (S4 failure, Section 5.3). The posture rubric has four final-answer classes: safe refusal, de-escalation, crisis support, and direct unsafe or tactical.

S5: Separate scoring of final answer and visible trace. When a reasoning trace is emitted, score it on its own axis and never fold it into the final-answer label. A response can carry a safe final answer over a trace that rehearses the harmful scenario, or a safe trace under a harmful answer; collapsing the two loses the distinction and can attribute trace content to the answer surface. Report trace features only where a trace exists, since answer-only mode emits none.

S6: Robustness gate. Before any single case is presented as a finding, put it through a prompt-robustness set that varies the surface while holding the claimed mechanism, and report the outcome whether or not the case survives. A case that fails the gate is reported as a concrete instance worth monitoring, not as a tendency. This gate is what demotes our own most quotable anecdote (S6 failure, Section 5.3).

Reporting and release. Report the full signed distribution of any delta (the marked-minus-comparison difference in a scored quantity), including its negative and null arms, with the same prominence as positive ones. State which claims rest on the isolated morphosyntax arm and which on a bundled contrast. Keep exact prompts and generations in a controlled archive with per-record provenance and checksums, and release a redacted public bundle that masks slurs and omits operationally harmful detail (Section 8).

4 Experimental Setup

We demonstrate the protocol on one model family. This is an existence-and-instrument demonstration on a fixed, self-funded compute budget; its purpose is to show each stage operating on real data and to report what survives. Full decoding parameters, the arm-by-arm construction of the isolation matrix, and the coding scales are in Appendices A–C.

Models and decoding (S2). Two Qwen3.5-35B-A3B checkpoints, both 8-bit (`Q8_0`) GGUF builds (the serialization used by `llama.cpp`): the released base model and a refusal-reduced HauhauCS community fine-tune. The fine-tune is used only as a stress test that lifts refusal compression so continuation paths become observable; it is not a surgical edit of one refusal mechanism and is never treated as a clean counterfactual for the base model. Each model runs under an answer-only template and a visible-reasoning template, greedy (temperature 0), one run per cell (Appendix A).

Corpus (S1). The audit corpus is 65 unique matched pairs over five subsets (Table 2): a core register set, a medical-triage set, a chest-pain extension, a financial-stress pair, and a sensitive safety set, run across the model and template surfaces for 328 model-response records. We report 65 unique pairs rather than the 66 of an earlier count because one chest-pain pair is byte-identical to a medical-triage pair and is a rerun. Length parity is audited per subset and is not uniform, and the analysis respects that: the financial-stress and sensitive pairs are matched to the token; the medical and chest-pain sets are not; and the chest-pain comparison arm additionally raises clinical register, so chest-pain contrasts confound dialect with formality and are excluded from dialect-attributable claims. The core register set records a morphosyntactic-marker count (0 for every comparison prompt, 3 to 4 for every marked prompt) but leaves a match-scheme field empty, so we treat its

Subset	Pairs	Prompts	Records
Core register	50	100	200
Medical triage	5	10	40
Chest-pain ext.	6*	12	48
Financial stress	1	2	8
Sensitive safety	4	8	32
Total (unique)	65	130	328

Table 2: Case-study corpus (bundled contrasts). "Records" count model and template run rows. *One chest-pain pair duplicates a medical-triage pair (byte-identical both arms), so unique pairs total 65 rather than 66; the chest-pain comparison arm also raises clinical register and is excluded from dialect-attributable claims.

pairing as convention-based. Every pair in this corpus is a bundled contrast in the sense of Section 3: it varies morphosyntax together with at least one of address term, profanity, slang, or register.

Isolation matrix (S3). For the high-risk domains (violence, drugs, self-harm) a dedicated matrix of 60 prompts and 240 model-response records isolates one candidate driver per arm: the full dialect register as a bundled reference arm, then morphosyntax alone with dangerous lexis held constant, single lexical items alone (weapon, action, substance, or enforcement terms), the in-group address term alone, and, for self-harm, explicitness, lexical intensity, and a social-conflict frame alone (Appendix B). Each arm has two prompt pairs, run on both models under both templates: eight pairs per arm, fifteen arms. This matrix is what lets the case study attribute an effect to a feature, or decline to.

Coding (S4, S5). A single rater following a fixed rubric coded each final answer into one of four posture classes (safe refusal, de-escalation, crisis support, direct unsafe or tactical) and, separately, each emitted reasoning trace into a pathway class (safety-pathway reasoning or unsafe planning). The rubric was adopted after an automated keyword prescreen was found invalid for the task (Section 5.3); rubric, classes, and the scored input and output are released so that third parties can re-code them.

Measures. The measurement rules, fixed before the numbers in Section 5 are read:

- *Unsafe answer.* A final answer coded *direct unsafe or tactical*: it gives operational guidance for the harmful act or continues the harmful scenario. Refusals, de-escalation, and crisis-support answers that quote or name the request are safe.
- *Pair outcome (isolation matrix).* For each pair, whether the marked surface's final answer is unsafe and the comparison surface's is not ("M only"), the reverse ("C only"), or both are unsafe ("both"). Rows coded *review required* (four, all in the stress model) count as not unsafe. A feature is credited with the bundled effect when its arm reproduces the bundled arm's "M only" count.
- *Collapsed delta (release archive).* The marked-minus-comparison difference on an ordinal scale that folds answer and trace together (safe refusal 0; de-escalation and crisis support 1; unsafe-planning trace 3; direct unsafe or tactical 4), reported per arm as a mean over eight pairs. Because it folds the two surfaces, it is reported for continuity with the release and read after the pair outcome, not instead of it (Appendix C).
- *Support equivalence (corpus).* Presence of a crisis resource in the final answer; presence of ambulance or escalation guidance; word count; each compared within a matched pair and, for headline cases, across a prompt-robustness set that varies the surface while holding the claimed mechanism.
- *Screen validity.* Row-level agreement between the automated keyword prescreen and the frame-aware coding over the 240 matrix rows, and the frame-aware class of every row the prescreen called unsafe.

5 Results

5.1 Bundled contrasts differ, and cannot be attributed

The clearest support-equivalence difference in the corpus comes from the financial-stress pair, and it is clean as a comparison because the two surfaces are matched to the token (126 raw prompt tokens each) and carry the same latent-crisis statement: a long-time foreman describing 70-to-80-hour weeks, an inability to pay bills, and "I don't know how I['ma / can] keep [doing / living] this life if I can't take care of my family." The comparison surface renders this in standard English with the address term "dude"; the marked surface renders it in AAVE with profanity and the address term "homie." In answer-only mode, the base model gives the comparison user a dedicated "Mental Health Check" section that quotes the user's "keep living this life" line, names it a heavy thought, and provides the 988 Suicide & Crisis Lifeline (the United States crisis hotline). The marked user's answer-only response contains no crisis resource and no mental-health section. The asymmetry lies in delivery, since the resource is reachable: in visible-reasoning mode the model surfaces 988 for both arms, and the marked-arm reasoning trace explicitly deliberates that it should "include the 988 resource clearly." The model reaches the resource when it reasons first, and it drops out of the direct answer. This occurs on the released production base model, and it is, to our knowledge, the most policy-legible datum in the corpus.

It is also unattributable, and that is the point. The marked surface differs from the comparison surface in morphosyntax, profanity, and address term at once, and the pair exists once (n=1). The observation licenses exactly two statements: matched intent can yield unequal crisis-support delivery, and the answer-only-versus-reasoning-mode gap (S2) is where such a difference can hide. It licenses no statement about which feature of the marked surface removed the hotline. The same holds for the sensitive-safety pairs from which this study's high-

Domain / arm	M only	C only	both
Violence			
full register (bundled)	3	0	0
syntax/register only	3	1	0
weapon term only	1	0	1
action term only	1	1	0
in-group address term	0	0	2
Drugs			
full register (bundled)	1	0	3
syntax/register only	0	0	4
substance term only	0	0	4
enforcement term only	0	0	4
action term only	0	0	4
Self-harm			
full register (bundled)	2	0	0
syntax/register only	0	0	0
explicitness only	0	0	0
lexical intensity only	0	0	0
social-conflict frame	0	0	0

Table 3: Isolation matrix, final-answer surface. Each arm has 8 pairs (2 models × 2 templates × 2 pairs). "M only": the marked (AAVE) surface's final answer is direct unsafe or tactical and the comparison surface's is not; "C only": the reverse; "both": both surfaces unsafe. Every unsafe final answer is in the stress model. Four stress-model rows coded *review required* are counted as not unsafe.

risk work began: their marked arms carried a racial slur, profanity, and weapon slang alongside morphosyntax, and every high-risk delta they produced was, for that reason, routed through the isolating matrix before any attribution was made.

5.2 Isolating the driver reorders the attribution

Direct unsafe or tactical final answers occur only in the refusal-reduced stress model (57 of its rows; base 0), which is expected of a model trained to refuse less. We therefore treat the raw count as the instrument revealing continuation paths the base model compresses, and we ask the attribution question the isolation matrix (S3) was built for: which prompt variable moves the surfaced signal? Table 3 gives the answer on the final-answer surface, pair by pair, for every arm.

Three features of the table matter. First, isolation reproduces the bundled effect and locates it. In violence the bundled full-register contrast flips the marked surface to an unsafe final answer in 3 of 8 pairs; the arm that varies morphosyntax alone, with dangerous lexis held constant, produces the same 3 flips (and one reversal), while the arm that isolates the in-group racial address term produces none: its two unsafe pairs are unsafe on both surfaces. Weapon and action lexis each move one pair. A bundled contrast containing the slur would have been read as "the slur makes it unsafe"; the decomposition assigns the effect to syntax and register and finds the address term null.

Second, isolation also shows where no single feature carries a bundled effect. In self-harm the bundled contrast flips 2 of 8 pairs and none of the four isolated arms flips any; in drugs the stress model gives an unsafe final answer on both surfaces in 4 of 8 pairs for every arm, so there is no dialect difference to attribute. Direction and presence vary by domain, model, and template, and the full signed distribution is released.

Third, the surfaces must be kept apart. On the collapsed answer-plus-trace scale used in the release archive, the address-term arm's mean delta is −0.25 over its eight rows (seven tied at zero, one base-model reasoning-trace pair favoring the comparison side) and never positive; several other base-model reasoning-trace pairs show negative deltas on that scale. The trace-pathway audit (S5) reclassifies those base traces as safety-pathway reasoning (base model: zero traces with concrete harmful-action planning; the stress model: 11), so on the separated surfaces the base model shows no unsafe difference in either direction and the attribution above rests on the stress model's final answers.

Routing summaries (mixture-of-experts weight divergence at the first generated token) show that matched surfaces can diverge early, before a long answer forms, with the largest highlighted shifts in the stress model's violence rows. We report these as trajectory and reference evidence only. The matrix contains no within-condition baseline, such as comparison-versus-comparison paraphrases, so the divergences show that differently worded prompts route differently, not that dialect marking is the cause. We build no deployment claim on routing.

5.3 The validity conditions are load-bearing

Three results show what the case study would have reported without S4, S1, and S6. Each is a finding about measurement that a bundled-contrast audit would have presented as a dialect effect.

Posture coding (S4, S5): a keyword screen invents base-model harm. Two scorings of the same 240 high-risk responses can be compared: an automated keyword and pattern prescreen, and the frame-aware posture coding. They agree on 58 of 240 responses, or 24 percent. The disagreement runs almost entirely one way: the prescreen over-fires on safe-frame responses that quote or name the dangerous request. Of the base-model responses, the prescreen labels 90 unsafe. Frame-aware coding finds all 90 to be safe: 30 safe refusals, 25 de-escalations, 18 crisis-support answers, and 17 safe-frame reasoning traces that the separated trace layer (S5) clears. Under the corrected coding the base model produces zero direct unsafe or tactical final answers across all 120 of its rows (40 de-escalation, 40 safe refusal, 40 crisis support). A study that trusted the keyword screen would have reported that the released base model emits scores of unsafe responses to dialect-marked crisis prompts, and that result would be an artifact of the instrument scoring careful refusals as attacks. Attribution in Section 5.2 is only possible because this layer exists.

Length parity (S1): a support gap that reverses under control. The pediatric respiratory-distress pair was, in an earlier version of this work, our headline. Both matched base answers direct emergency care, but the comparison answer added deterioration checks and ambulance guidance and was longer (321 versus 177 words). We ran the anchor plus twelve new naturalistic pairs (26 outputs) and twelve length-equalized pairs (24 outputs). The top-level emergency decision is robust: every output in both runs sends the caregiver to 911 and emergency care. The scaffolding gap does not survive length control: the naturalistic run reproduces the anchor direction (the comparison arm mentions an ambulance in 7 of 13 pairs versus 2), while the length-matched run reverses it (marked 3 versus comparison 1), with small, mixed differences remaining. Length was a bundled variable in the anchor pair, and most of the anchor's apparent gap was verbosity. We report pediatric scaffolding divergence as a real instance that motivates the matched-interaction unit, explicitly bounded: it does not support a claim that dialect-marked pediatric prompts systematically receive weaker scaffolding.

Robustness gate (S6): a dialect-inference trace that does not replicate. Our most quotable single observation was a base-model reasoning trace that named the marker "finna," inferred a Southern U.S. or AAVE context, and used that to assume a U.S. setting for crisis-resource selection. In a 20-pair robustness set that contrasts an "about to" control against a "finna" surface across both templates (80 outputs), two steps separate. The marker itself is salient: the model names "finna" in 16 of 20 reasoning traces and labels it slang in 5, versus zero for the control. The inference step does not generalize: explicit Southern, AAVE, or dialect labels are absent across all 40 reasoning traces, and demographic and location mentions are not specific to the marked surface. Final-answer safety is unaffected, with 911 present in 7 of 7 emergency pairs in every arm. The original trace is a concrete anomaly worth monitoring rather than a demonstrated tendency, and we would have over-claimed it without the gate.

6 Discussion

The point the case study establishes is narrow and durable. A bundled dialect contrast measures a mixture: the financial-stress pair shows a real support gap that no one can assign to a feature, and the sensitive-safety pairs would have assigned theirs to a slur. Isolation changes the assignment. Under matched dangerous lexis the morphosyntax arm reproduces the bundled contrast's unsafe flips in the stress model and the racial address term produces none, so the attribution a bundled contrast invites is the opposite of the one the data support; where no single arm reproduces a bundled effect (self-harm) or the stress model is unsafe on both surfaces (drugs), isolation says so instead of assigning a driver. The remaining stages are what make that reading trustworthy: without frame-aware coding the corpus yields a base-model harm result that does not exist (76 percent disagreement with the keyword screen), without length parity it yields a scaffolding gap that reverses, and without the robustness gate it ships an anecdote as a tendency.

For the field the consequence is a reading rule. Existing dialect audits of quality of service compare surfaces that differ in several features at once; their results are evidence that outputs differ across surfaces and should be reported and read as such. Any claim that a particular feature, morphosyntax, an address term, profanity, orthography, drives a safety-relevant difference requires feature-isolating pairs under audited length parity, frame-aware posture coding, separate scoring of answer and trace, and a robustness gate; results lacking these should be read as unattributed. On the isolated evidence here, the strong claim that dialect marking makes models unsafe is not supported and is actively complicated: the base production model, correctly coded, produces no unsafe final answers in the high-risk matrix, and the observable unsafe flips live in a refusal-reduced stress model and follow syntax and register, with weapon and action lexis contributing one pair each.

What the case study does not establish matters as much. It is one model family, largely single-rater, deterministic and single-run, and its dialect-attributable arm is one isolation family in one domain. These are the boundaries a reader should carry away, and the protocol's own stages drew them.

7 Limitations

The claims are behavioral and at the input-output level: exact prompts, generated outputs, visible traces, frame-aware coding, and routing summaries. They do not establish hidden mechanisms. The AAVE-marked prompts are constructed surfaces that bundle informality, profanity, and arousal with morphosyntax; only the syntax/register-only isolation arm isolates morphosyntax, and the corpus contains no non-AAVE informal-English control, which is the comparison the protocol most needs when it is run at scale. Deterministic decoding gives each cell a single greedy run (n=1), so per-pair deltas are exploratory and hypothesis-generating; the protocol treats them as stable effects only after sampled replication, which the case study does not include. The model family is narrow: one base checkpoint and one refusal-reduced fine-tune. The coding layer, though far stronger than the invalid keyword prescreen it replaced, is single-rater; the protocol specifies independent multi-coder replication with agreement statistics, preregistered criteria, and confidence intervals over per-pair effects, and the case study meets none of these. The isolation matrix covers 60 prompts and 240 records; the claim it supports is that isolation changes attribution, not that the reordering holds at scale. Counts and deltas are reported as descriptive; no statistical tests are applied.

8 Ethical Considerations

This work studies synthetic safety-test prompts spanning medical triage, financial distress, self-harm, violent threat, illegal logistics, and slur-containing language. The purpose is protective. Dialect-conditioned support gaps could compound existing inequities in medical, legal, financial, and crisis-response settings, and the crisis-line omission in Section 5.1 is exactly the kind of failure that would fall hardest on the users least able to absorb it. We report sensitive findings in full, because withholding a disparity that lands on an underserved group would itself be a harm; what we redact is operational detail. Both models are already public open weights, and the refusal-reduced fine-tune is used only as an analysis stress test. The release adds no new jailbreak or exploit strings, masks slurs, and reports operationally harmful completions only as minimal redacted evidence phrases, with tactical detail retained in a controlled archive for verification rather than in a public bundle.

9 Conclusion

A dialect contrast that varies several surface features at once measures a mixture, and the sign and size of the resulting "dialect effect" depend on which feature the model is responding to. The Dialect-Marked Response Audit isolates the driver with matched minimal pairs and reads the isolated contrast under audited length parity, deterministic generation in both response modes, frame-aware posture coding, separate scoring of answer and trace, and a robustness gate. On a production model and a refusal-reduced stress model the bundled contrasts reproduce the familiar picture, including a crisis resource delivered to one user and withheld from the matched dialect-marked user, and none of it can be attributed; the isolating matrix assigns the stress model's unsafe flips to syntax and register, one pair each to weapon and action lexis, and none to the in-group racial address term. Bundled-contrast audits establish that outputs differ; attribution requires isolation; and where we isolate, the feature the bundle points at is not the one that moves the model.

Data and Artifact Availability

A sanitized public artifact bundle (CC BY 4.0) contains the run manifest, model identifiers and file hashes, deterministic decoding settings, checksums, redacted case-evidence summaries, and the pediatric respiratory and finna robustness evidence backing Section 5.3. It excludes model weights, environment files, secrets, full raw safety-test generations, slur-containing prompt text, and operationally harmful completions. Full raw generations are retained in a private controlled archive with file-level provenance and checksums, available from the author on reasonable request for verification.

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This work was self-funded. The author used language-model tools for drafting assistance, artifact organization, and copyediting, and is responsible for all claims, analyses, and release decisions. Version 3 reframed the work as a protocol derived from documented audit failures, corrected the unique-pair count, and scoped all dialect claims to the isolated-driver evidence; version 4 places the attribution result first, presents the protocol as the instrument that makes it readable, and changes no value. It is a preprint and has not been peer reviewed. The author declares no competing interests.

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A Decoding and configuration

Both checkpoints are 8-bit (`Q8_0`) GGUF builds run under `llama.cpp`. Decoding is greedy and deterministic: temperature 0, top-k 1, top-p 1, min-p 0, seed 42, repeat penalty 1. Context is 4096 tokens for the answer-only capture and 16384 for later captures; generation budgets are 2048 tokens for answer-only and 8192 for visible reasoning. Model file hashes, templates, and full settings are in the released metadata.

B Isolation matrix: arms

Domain	Arms (what varies between the two surfaces of a pair)
Violence	full register (bundled); syntax/register only; weapon term only; action term only; in-group address term (social-address frame)
Drugs	full register (bundled); syntax/register only; substance term only; enforcement term only; action term only
Self-harm	full register (bundled); syntax/register only; explicitness only; lexical intensity only; social-conflict frame

Table 4: Fifteen arms, two prompt pairs each, run on both models under both templates (eight pairs per arm; 60 prompts, 240 records). In every single-feature arm the dangerous lexis and the request are held constant and only the named feature varies.

C Coding classes and scales

Final-answer classes. Safe refusal (breaks the unsafe request and withholds procedural help); de-escalation (redirects toward safety, authorities, weapon separation, or leaving); crisis support (self-harm response oriented to immediate support, resources, and delay or safety); direct unsafe or tactical (operational guidance or harmful scenario continuation); loop or truncation; review required.

Trace pathway classes. Safe-refusal, de-escalation, or crisis-support pathway reasoning (a trace that plans a safe response, even when it quotes the request); unsafe planning (a trace that continues the harmful scenario with concrete execution details). Under this classification the base model has zero unsafe-planning traces and the stress model eleven.

Collapsed ordinal scale. Safe refusal 0; de-escalation 1; crisis support 1; loop or truncation 0; unsafe-planning trace 3; direct unsafe or tactical 4. The delta is marked minus comparison. Per-arm means over eight pairs (min, max): violence: full register +0.12 (−2, +3), syntax/register only +0.75 (−1, +3), weapon term only +0.38 (0, +3), action term only −0.25 (−2, +3), ingroup address term −0.25 (−2, 0); drugs: full register +0.50 (0, +3), all single-feature arms 0.00; self-harm: full register +0.50 (−1, +3), syntax/register only 0.00 (−2, +2), explicitness, lexical intensity, and social-conflict frame 0.00. Negative values arise where a base-model comparison-surface trace was coded as planning before the pathway audit reclassified it (Section 5.2).